Def. Let S be a regular surface and T_pS be the tangent plane at $p \in S$.
A quadratic form

$$I_p : T_pS \to \mathbb{R} : w \mapsto \langle w, w \rangle_p \geq 0$$

is called the first fundamental form of S at p.

Computation of the fundamental form.
Given $p \in S$, let $X : U \to S$ be a parametrization of S containing p.
Let $w \in T_pS$ be given. Take a curve

$$\alpha : (-\varepsilon, \varepsilon) \to X[U] : t \mapsto X(u(t), v(t)) \quad \text{such that } \alpha(0) = p \text{ and } \alpha'(0) = w.$$

Now,

$$\begin{aligned} I_p(w) = I_p(\alpha'(0)) &= \langle \alpha'(0), \alpha'(0) \rangle_p \\ &= \langle X_u u' + X_v v', X_u u' + X_v v' \rangle \\ &= \langle X_u, X_u \rangle \cdot (u')^2 + 2 \cdot \langle X_u, X_v \rangle \cdot u'v' + \langle X_v, X_v \rangle \cdot (v')^2 \quad (t = 0). \end{aligned}$$

In this sense, define

$$E(u_0, v_0) = \langle X_u, X_u \rangle_p, \quad F(u_0, v_0) = \langle X_u, X_v \rangle_p, \quad G(u_0, v_0) = \langle X_v, X_v \rangle_p.$$

Def. A length of parametrized curve $\alpha : I \to S$ is

$$s(t) = \int_0^t |\alpha'(r)| \, dr = \int_0^t \sqrt{I(\alpha'(r))} \, dr = \int_0^t \sqrt{E \cdot (u')^2 + 2F \cdot u'v' + G(v')^2} \, dr$$

Def. Let $R \subset S$ be a bounded set contained a image of $X : U \to S$,
A area of R is:

$$A(R) = \iint_{X^{-1}[R]} |X_u \times X_v| \, du dv.$$

Examples,

Let $S = \{(x,y,z) \mid x^2+y^2=1\}$ be the sylinder.

Consider a parametrization of S,

$$X : U \to \mathbb{R}^3 : (u,v) \mapsto (\cos u, \sin u, v)$$

where $U = \{(u,v) \in \mathbb{R}^2 \mid 0 < u < 2\pi, -\infty < v < \infty\}$,

Simply, $X_u = (-\sin u, \cos u, 0)$ and $X_v = (0, 0, 1)$.

So, $E = 1$, $F = 0$, $G = 1$.

Helicoid.

Let $S = \{(r \cdot \cos\theta, r \cdot \sin\theta, a$

Exercises. Sec 2.5.

1. Evaluate the first-fundamental form of the following:

a). $X(u,v) = (a\cdot\sin u\cdot\cos v,\ b\cdot\sin u\cdot\sin v,\ c\cdot\cos u)$

b). $X(u,v) = (a\cdot u\cos u,\ b\cdot u\sin v,\ u^2)$

Proof.

a), $X_u = (a\cos u\cdot\cos v,\ b\cos u\cdot\sin v,\ -c\cdot\sin u)$,

$X_v = (-a\cdot\sin u\cdot\sin v,\ b\cdot\sin u\cdot\cos v,\ 0)$.

So, $E = \langle X_u, X_u\rangle = a^2\cdot\cos^2 u\cdot\cos^2 v + b^2\cos^2 u\cdot\sin^2 v + c^2\cdot\sin^2 u$

$= (a^2\cos^2 v + b^2\sin^2 v)\cos^2 u + c^2\cdot\sin^2 u$

$F = \langle X_u, X_v\rangle = -a^2\cdot\cos u\cdot\sin u\cdot\cos v\cdot\sin v + b^2\cdot\cos u\cdot\sin u\cdot\cos v\cdot\sin v$

$= (-a^2+b^2)\cdot\cos u\cdot\sin u\cdot\cos v\cdot\sin v.$

$G = \langle X_v, X_v\rangle = a^2\cdot\sin^2 u\cdot\sin^2 v + b^2\cdot\sin^2 v\cdot\cos^2 v.$

Fundamental form of the Monge patch.

Given a differentiable map $f: \mathbb{R}^2 \to \mathbb{R}$, consider the Monge patch

$$X: U \to \mathbb{R}^3 : (u, v) \mapsto (u, v, f(u,v)).$$

Then, the partial derivative is

$$X_u = (1, 0, f_u) \text{ and } X_v = (0, 1, f_v)$$

so, the first-fundamental form is

$$E = \langle X_u, X_u \rangle = 1 + (f_u)^2, \quad F = \langle X_u, X_v \rangle = f_u \cdot f_v, \quad G = \langle X_v, X_v \rangle = 1 + (f_v)^2.$$

Moreover, $\det \begin{pmatrix} E & F \\ F & G \end{pmatrix} = 1 + (f_u)^2 + (f_v)^2 \geq 1 > 0$.

Notation

Denote the Coordinate Patch as

$$X: U \to \mathbb{R}^3 : (u^1, u^2) \mapsto (f^1, f^2, f^3)$$

Second fundamental form.

Def. Let $X: \mathcal{U} \to \mathbb{R}^3$ be a coordinate patch.
The second fundamental form of X is:

$$L_{ij} \overset{\text{def}}{=} \langle X_{ij}, n \rangle \quad (1 \le i, j \le 2)$$

And the Christoffel symbols is

$$\Gamma_{ij}^{k} \overset{\text{def}}{=} \sum_{l=1}^{2} \langle X_{ij}, X_l \rangle \cdot g^{lk}.$$

where $X_{ij} := \dfrac{\partial^2 X}{\partial u^i \partial u^j}$ and g^{lk} is (k,l) entry of $\begin{pmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{pmatrix}^{-1}$.

$$= \begin{pmatrix} g_{22} & -g_{12} \\ -g_{21} & g_{11} \end{pmatrix}.$$

$$n = \frac{X_1 \times X_2}{\| X_1 \times X_2 \|}.$$

Normal and geodesic Curvature.

Def Let $X: U \to \mathbb{R}^3$ be a patch and

$$\gamma: I \to \mathbb{R}^3 : s \mapsto X(\gamma^1(s), \gamma^2(s))$$

be a unit speed curve whose image is contained in the patch.

Define the Normal Curvature

$$k_n(s) = \langle \gamma''(s), n(s) \rangle$$

and the Geodesic Curvature

$$k_g(s) = \langle \gamma''(s), n(s) \times T(s) \rangle$$

So that $k(s) \cdot N(s) = T'(s) = \gamma''(s) = k_n(s) \cdot n(s) + k_g(s) \cdot n(s) \times T(s)$.

And $k^2 = k_n^2 + k_g^2$

Normal of Monge patch.

Let $X: U \to \mathbb{R}^3 : (u, v) \mapsto (u, v, f(u,v))$

$$\begin{matrix} 1 & 0 & f_u \\ 0 & 1 & f_v \end{matrix}$$

Then,

$$n = \frac{X_u \times X_v}{\| X_u \times X_v \|} = \frac{(-f_u, -f_v, 1)}{\sqrt{1 + f_u^2 + f_v^2}}$$

Example.

Consider the upper hemisphere

$$X : \mathbb{D} \to \mathbb{R}^3 : (u, v) \mapsto (u, v, \sqrt{1-u^2-v^2})$$

Let a unit speed curve in X

$$\gamma : (-\tfrac{\pi}{2}, \tfrac{\pi}{2}) \to X[\mathbb{D}] : t \mapsto (\sin t, 0, \cos t).$$

The Frenet frame is

$T(t) = (\cos t, 0, -\sin t)$ $\quad \kappa(s) = 1$

$N(t) = (-\sin t, 0, -\cos t)$ $\quad \tau(s) = 0.$

$B(t) = (0, 1, 0).$

$f_u = \frac{-2u}{2 \cdot f} = -\frac{1}{f}$, $f_v = -\frac{1}{f}$.

$f_u = -u/f$, $f_v = -v/f$

And the normal

$$n(t) = (u/f, v/f, 1) \Big/ \underbrace{\sqrt{1 + \frac{u^2+v^2}{f^2}}}_{= f}$$

$$1 + \frac{u^2+v^2}{f^2} = \frac{1-u^2-v^2}{f^2}$$

$$= (u, v, f).$$

The normal curvature at $\gamma(t) = (\sin t, 0, \cos t)$ is

$$k_n = \langle T', n \rangle = \langle (-\sin t, 0, -\cos t), (\sin t, 0, \cos t) \rangle = -1.$$

But, since the curvature $k^2 = k_n^2 + k_g^2$, so $k_g = 0$.